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Installation survey

Hydraulic system single-circuit

Chassis Model	Sales designation	Installation
435.115	U 1300 L	SA 35 754 with SA 35 759
.117	U 1300 L/37	
.110	U 1700	SA 35 847
.111	U 1700 L	SA 35 920

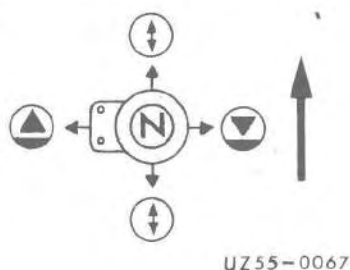
Hydraulic system dual-circuit

Chassis Model	Sales designation	Installation
435.115	U 1300 L	SA 36 012 with SA 35 847 SA 36 083
.110	U 1700	
.111	U 1700 L	

Hydraulic control equipment

Version	Single-circuit system	Dual-circuit system
One-cell	E1 SZ	—
Two-cell	E1 VZ SZ	—
Three-cell	E1 VZ VZ SZ	—
Four-cell	E1 VZ VZ VZ SZ	E1 VZ VZ E2 SZ

E1 Inlet plate
E2 Inlet plate for circuit 2
VZ Valve cell
SZ End plate



Symbols for actuating levers

Shift positions

- Floating position
- Position — Lower
- Neutral position
- Position — Raise

Hydraulic pump

		Single-circuit system		Dual-circuit system ¹⁾	
		1st version up to chassis end No. 080 146	2nd version from chassis end No. 080 147	Circuit 1	Circuit2
Version		Inner gear	Inner gear	Tandem hydr. gear pump	
Type		IPM 16645	IPSF-3-16644	HY/ZFFS 15	
Delivery volume	cm ³	15.8	16.0	8 5.5 2)	11 14 2)
Delivery flow at rated engine speed	l/min	53	53	25 17 2)	35 44 2)
Max. pressure P max.	bar	200 + 15			
Ratio engine speed/pump speed	i	1 : 1.25			
Belt pulleys dia.	mm	153			

- 1) Flow of circuit 2 can be diverted into circuit 1 via rotary knob valve.
2) Different circuit layout from 15.4.8.2

Capacities

	Service product	SAE grade	Zone	Capacity
— when refilling	Engine oil HD 226, 227 or hydraulic oil 341	10 W	Temperate zone	35
		5 W 20	Cold zone	
		30	Hot zone	
— when changing oil		1)	—	31
Permissible extraction volume		—		26

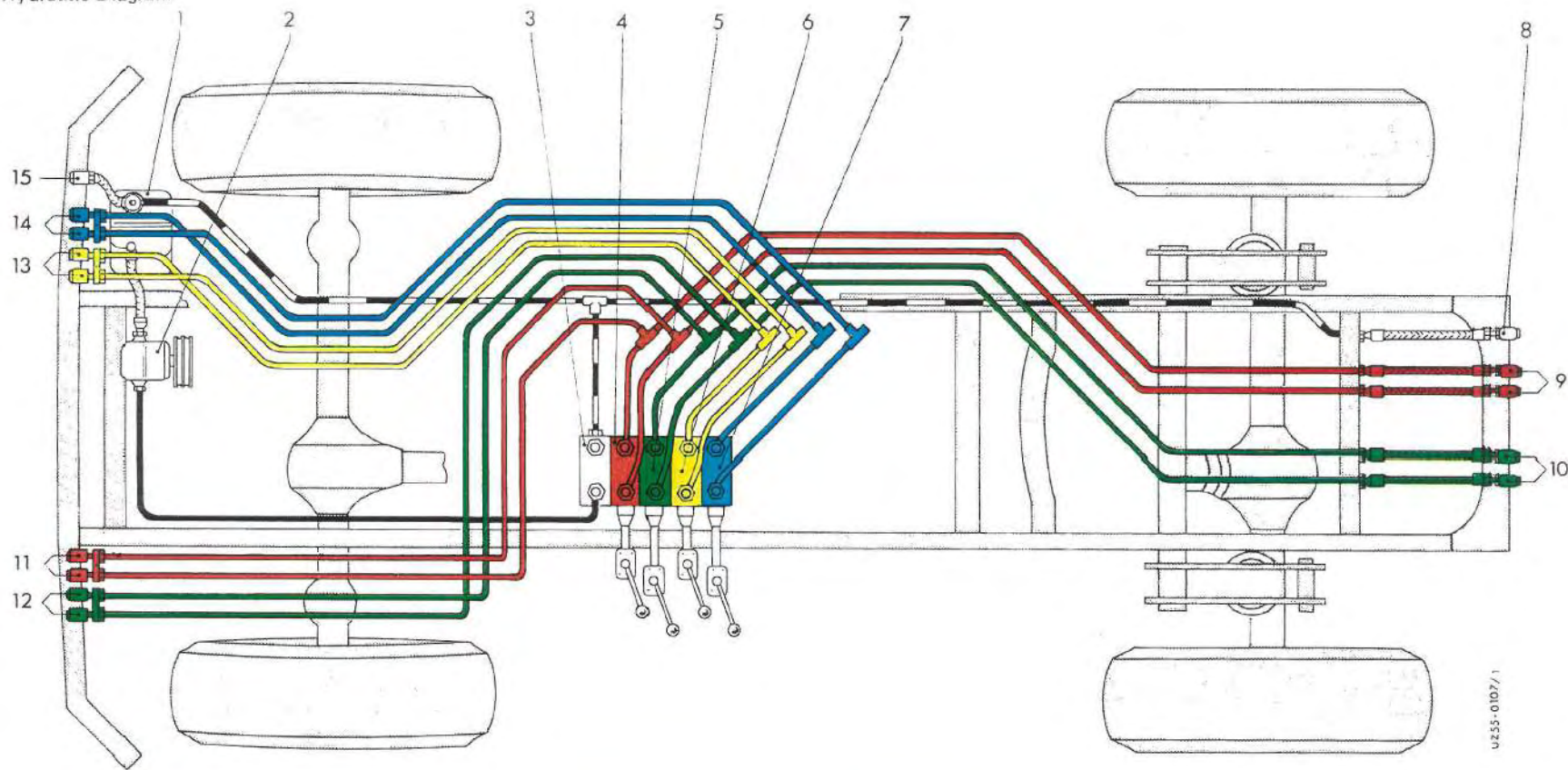
- 1) SAE grade as for refilling

Tilt cylinder

No. of stages		4
Lift height	mm	975
Nominal pressure	bar	200
Lift volume	l	5.7
Lifting force cardan ring	N	80 000

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Hydraulic Diagram

A —B —C —D —E —F —

Single-circuit hydraulic system

- 1 Oil reservoir
- 2 Hydraulic pump
- 3 Inlet plate
- 4 Valve cell 1, red
- 5 Valve cell 2, green
- 6 Valve cell 3, yellow
- 7 Valve cell 4, blue

- 8 Separator return, SR
- 9 Hydraulic connection, valve cell 1
- 10 Hydraulic connection, valve cell 2
- 11 Hydraulic connection, valve cell 1
- 12 Hydraulic connection, valve cell 2
- 13 Hydraulic connection, valve cell 3
- 14 Hydraulic connection, valve cell 4

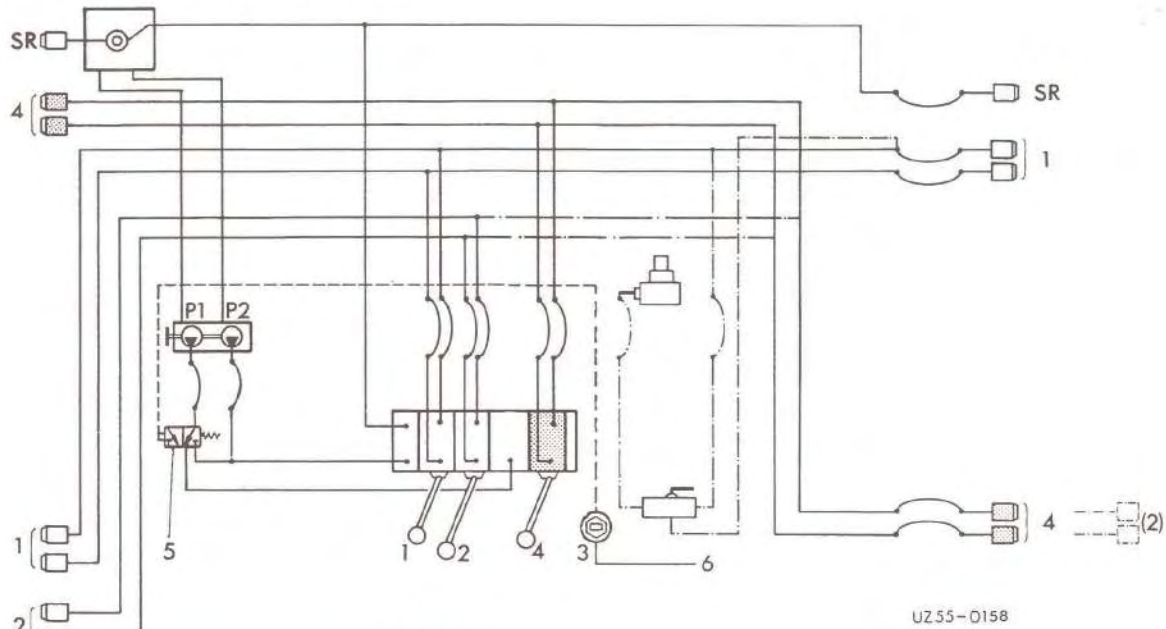
- 15 Separator return SR
- A Hydraulic circuit 1
- B Hydraulic circuit 2
- C Hydraulic circuit 3
- D Hydraulic circuit 4
- E Pressure line
- F Separate return line

SA 35 754

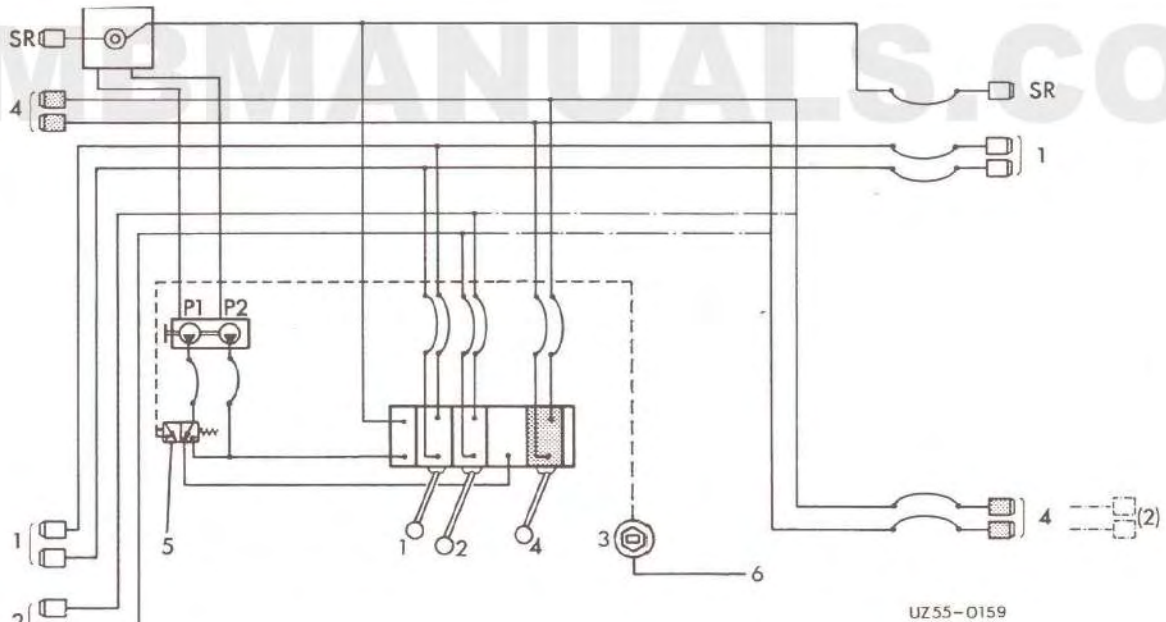
SA 36 012

Hydraulic diagram

Dual-circuit hydraulic system 435.110



Dual-circuit hydraulic system 435.111/115/117



Key

□ Circuit 1

■ Circuit 2

- 1 Hydraulic connections valve cell 1 and 2 1st circuit
- 2 Hydraulic connections valve cell 4 2nd circuit
- 3 Rotary knob valve (pneumatic)
- 5 Changeover valve 1st circuit/2nd circuit controlled by rotary knob valve

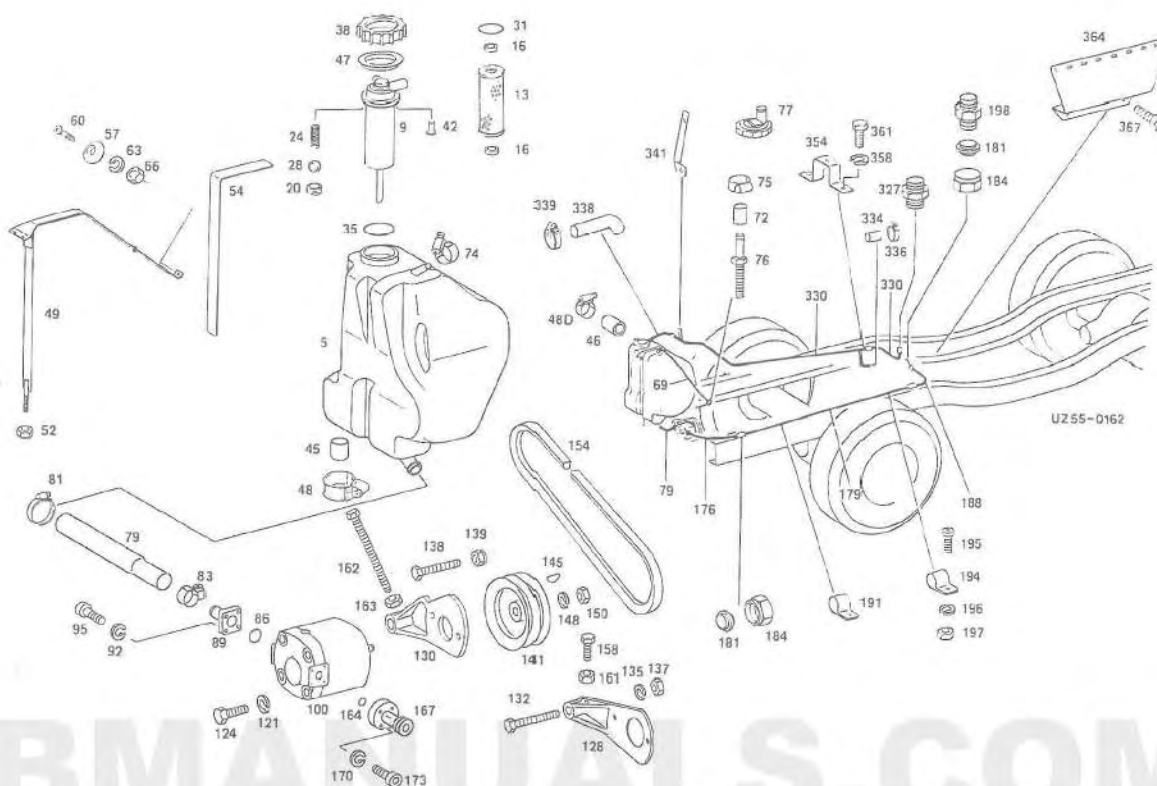
- 6 Line to 4-circuit protection valve
- P₁ + P₂ Tandem hydraulic pump
- SR Separate return line

Troubleshooting**Operating faults**

Leaks should immediately be rectified with the engine not running and the system not under load otherwise the entire hydraulic system will fail.

Fault	Cause	Remedy
Noises in hydraulic system.	V-belt for hydraulic pump loose.	Check V-belt and retension, replace if necessary.
	Oil level in reservoir too low.	Top up oil to prescribed level.
	Pump intake line leaking.	Check connections of intake line, tightening as necessary. Check intake hose, exchanging if necessary.
Pressure limiting valve chatters.	Air in hydraulic system.	Bleed system, examine intake line.
Hydraulic oil foams.	Intake line leaking, porous.	Check connections of intake line, tightening if necessary. Check intake hose, exchanging if necessary.
Hydraulic oil overheats.	Pressure limiting valve operates too soon.	Check pressure with gauge in socket, setting pressure limiting valve if need be or exchanging if necessary.
Attached unit is not fully raised	Insufficient oil in reservoir.	Top oil to prescribed level.
	V-belt loose.	Retension V-belt.
	Intake line leaking or damaged.	Tighten connections, exchange intake line if necessary.
	Insufficient pressure in system, pressure limiting valve operates too soon.	Check pressure with gauge in socket, setting pressure limiting valve if need be or replacing if necessary.
Control lever on hydraulic directional control valve does not operate in centre position.	Pressure limiting valve is not correctly adjusted.	Check pressure with gauge in socket, adjust pressure limiting valve if need be, replace as necessary.
	Valve spool jammed.	Fit new hydr. directional control valve.
Directional control valve leaking.	Pipe connections loose.	Tighten pipe connections.
	Control shaft of hydr. directional control valve leaking.	Replace sealing rings.
	Valve spool leaking.	Replace sealing ring.
	Screwplugs leaking.	Tighten screwplugs, replace sealing ring.
Hydraulic pump leaking.	Sealing rings on drive shaft leaking or damaged.	Replace sealing rings.

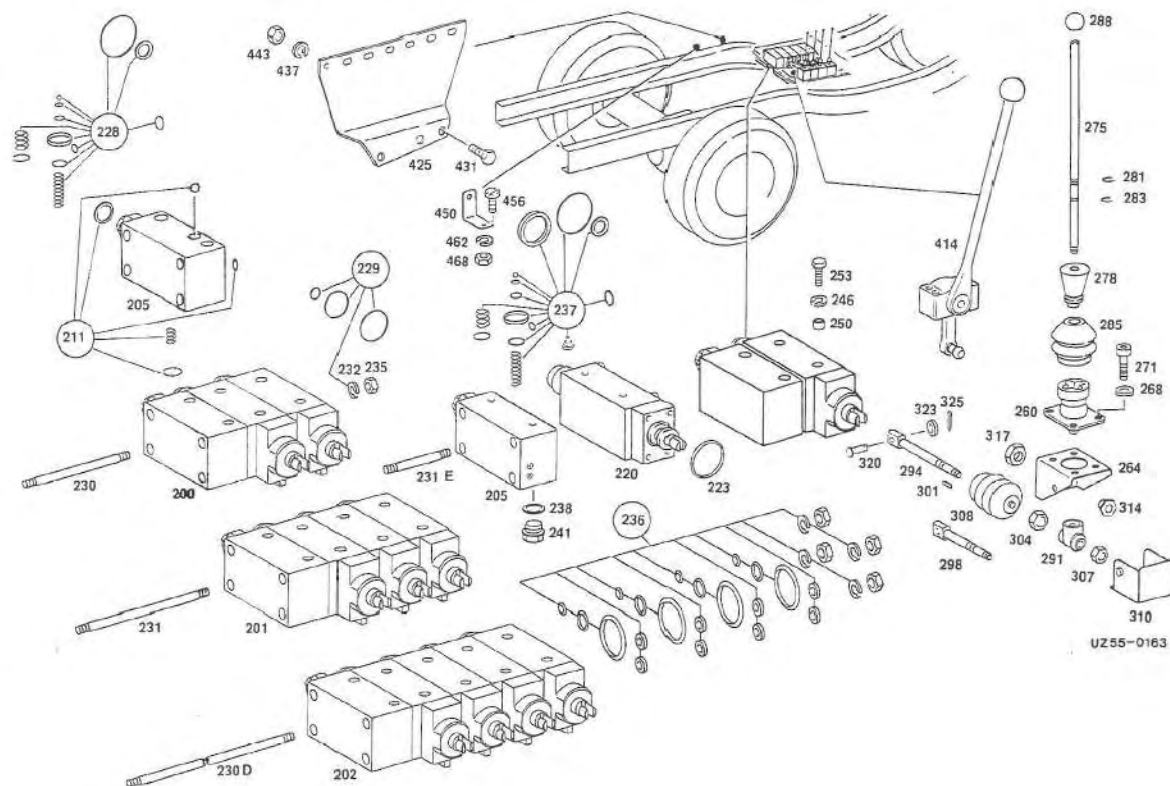
Exploded view



Hydraulic system

5	Oil reservoir	75	Clip	164	Sealing ring
9	Filter	76	Union	167	Flange
13	Filter element	77	Cover	170	Spring lock washer
16	Gasket	79	Rubber sleeve	173	Bolt
20	Valve seat ring	81	Clip	176	Hose line
24	Spring	83	Clip	179	Tube
28	Ball	86	Sealing ring	181	Cutting ring
31	Gasket	89	Pipe	184	Union nut
38	Nut	92	Spring lock washer	188	Hose line
42	Union	95	Bolt	191	Clip
45	Closing cover	100	Pump	194	Clip
46	Closing cover	121	Spring lock washer	198	Screw-in union
47	Ring	124	Bolt	327	Screw-in union
48	Clip	128	Bracket	330	Tube
48D	Clip	130	Bracket	334	Closing cover
49	Clamping strap	132	Bolt	336	Clip
52	Nut	135	Spring lock washer	338	Rubber sleeve
54	Base	137	Nut	339	Clip
57	Washer	141	Belt pulley	341	Bracket
60	Bolt	145	Woodruff key	354	Bracket
63	Spring lock washer	148	Spring washer	358	Spring washer
66	Nut	150	Nut	361	Bolt
69	Rubber sleeve	154	V-belt	364	Bracket
72	Hose	158	Bolt	367	Bolt
74	Clip	161	Nut		
		162	Bolt		
		163	Nut		

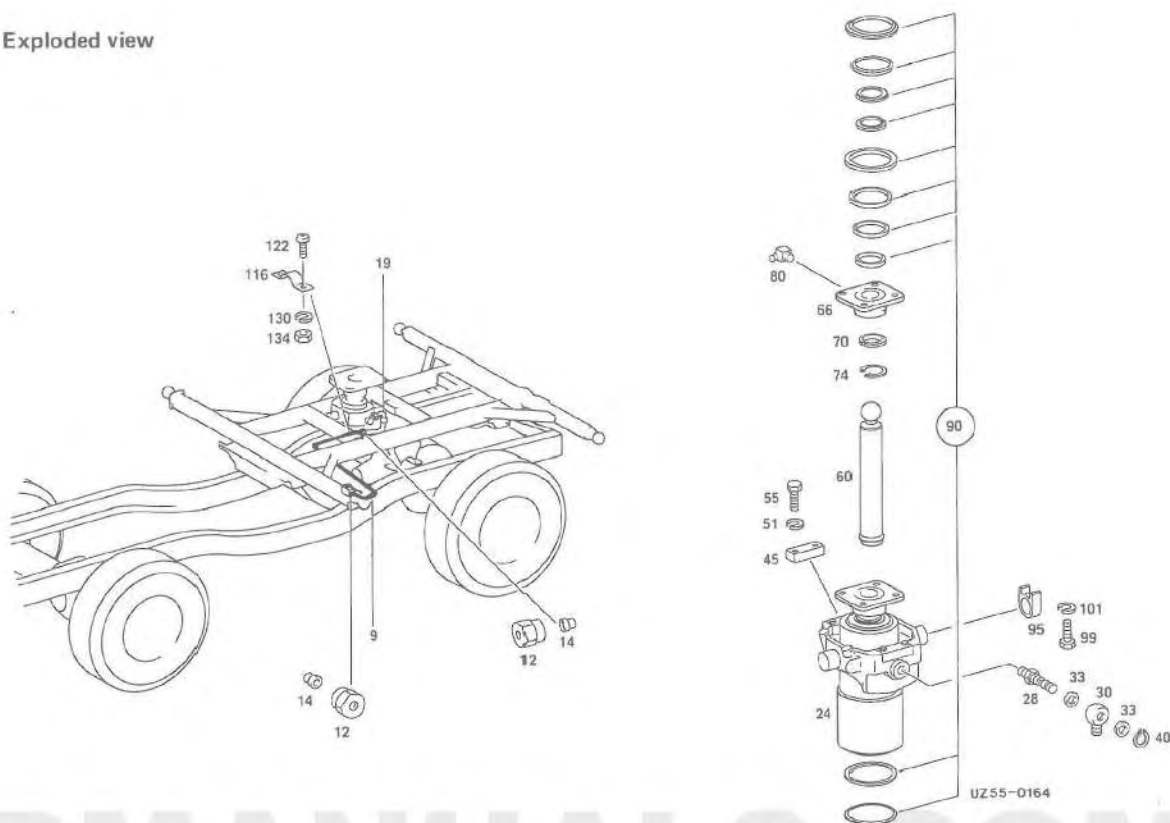
Exploded view



Hydraulic system

200	Control unit	246	Spring washer	304	Nut
201	Control unit	250	Spacer tube	307	Nut
202	Control unit	253	Bolt	308	Sleeve
205	Control unit	260	Housing	310	Cover plate
211	Repair set	264	Guide piece	314	Bush
220	Control unit	268	Spring washer	317	Nut
223	Sealing cord	271	Bolt	320	Pin
230	Stud	275	Lever	323	Washer
230D	Stud	278	Bush	325	Cotter pin
231	Stud	281	Snap ring	414	Shift lever
231E	Stud	283	Snap ring	425	Bracket
232	Spring lock washer	285	Sleeve	431	Bolt
235	Nut	288	Knob	437	Spring lock washer
236	Repair set	291	Guide piece	443	Nut
237	Repair set	294	Rod	450	Bracket
238	Sealing ring	298	Rod	462	Spring lock washer
241	Screw plug	301	Parallel key	468	Nut

Exploded view



Telescopic cylinder — Tipper

9	Tube	70	Spacer ring
12	Union nut	74	Circlip
14	Cutting ring	80	Grease nipple
19	Hose line	90	Repair set
24	Cylinder	95	Bush
28	Screw-in union	99	Bolt
30	Housing	101	Spring washer
40	Circlip	116	Clip
45	Web	122	Bolt
51	Spring washer	130	Spring washer
55	Bolt	134	Nut
60	Piston		
66	Ball socket		